

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 — SUB-STRAND 1.3 NUTRITION IN ANIMALS

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the unit. Use everything you learned from the Ugali and Sukuma Wiki Puzzle: your food journey maps, the Mandazi Moment, the enzyme charts, the food tests, the structure diagrams and your 3D model. Write your answers in the space under each prompt. Use full sentences and scientific words such as enzyme, substrate, mechanical digestion, chemical digestion, villi and sphincter. Always link your ideas to the school lunch of ugali, sukuma wiki and nyama choma. Where a question asks you to use evidence, say which test or observation you are using.

Part 1: Two ways of breaking down the meal

Think of the Kenyan school lunch of ugali, sukuma wiki and nyama choma. (a) Explain the difference between mechanical digestion and chemical digestion. (b) Give one example of each from the mouth or the stomach. (c) During the Mandazi Moment you chewed a piece of mandazi for 60 seconds. Describe what you noticed about its taste and explain what this tells you about what happens to the ugali in your mouth. (d) Would you expect the nyama choma to change taste in the mouth in the same way? Explain your answer.

Write your answer here:

Part 2: How does the body know which chemical to release?

One of the puzzles in Lesson 1 was how the body seems to 'know' which digestive chemical to use at each stage. (a) Explain how the digestive system manages to have the right chemical in the right place. (b) Name the enzyme that acts on each of these in the school lunch and the region where it acts: starch in ugali, protein in nyama choma, and fat in cooking oil. (c) Look at your food test results from Lessons 4 and 5. Name the test and the colour you saw for at least two nutrients, and say what these results tell you about the nutrients in the meal.

Write your answer here:

Part 3: Why does digestion take different times?

Another puzzle was why ugali seems to be digested faster than nyama choma. (a) Using what you learned about the regions of the alimentary canal, compare how and where the starch in ugali and the protein in nyama choma are digested. (b) The cooking oil and butter in the meal contain fat. Explain why fat digestion needs an extra step.

Write your answer here:

Part 4: Structures that make absorption and protection possible

(a) The small intestine is a tube only about 3 cm wide, yet it absorbs the nutrients from the whole meal. Explain how its structure makes this possible. Name the nutrients from ugali and nyama choma that are absorbed. **(b)** Explain how the stomach avoids digesting itself even though it contains acid and pepsin. **(c)** Describe the job of the cardiac sphincter and of the pyloric sphincter.

Write your answer here:

Part 5: Answering the driving question

Driving question: How is the human digestive system adapted to break down and absorb the nutrients in a typical Kenyan meal — and what happens when those adaptations fail? (a) Write a connected explanation that follows ugali, sukuma wiki and nyama choma through the alimentary canal. Include at least one mechanical process, at least three enzymes with their regions, and one structural adaptation. (b) Choose ONE adaptation from this unit and predict what would happen to the digestion or absorption of the meal if it failed. (c) In the Benedict's test on the simulated digested-ugali sample (a glucose solution), what did you observe? Explain what this result tells you about reducing sugars and digestion, and say how it relates to what happens to ugali starch in the body. (d) Compare your Lesson 1 food journey map with your 3D model. Name two things you can now explain that you could not explain at the start.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Mechanical and chemical digestion	Clearly distinguishes the two types with correct examples and uses the Mandazi Moment evidence to explain amylase action on starch.	Distinguishes the two types with correct examples but gives only a weak link to the Mandazi evidence.	Mixes up the two types or gives no examples or evidence.
Enzymes, specificity and food tests	Matches each enzyme to its substrate and region accurately, explains specificity, and uses named food tests with correct colours as evidence.	Matches most enzymes to substrates and regions and mentions some food tests.	Names few enzymes correctly or does not link enzymes to substrates or tests.
Structural adaptations and self-protection	Explains villi and microvilli, mucus and pepsinogen, and both sphincters accurately, linking each structure to its	Explains most structures and functions with minor gaps.	Names structures without explaining their functions, or confuses them.

	function in the meal.		
Connected explanation of the driving question	Gives a sequenced explanation of the meal through the canal and a reasoned prediction of what happens when an adaptation fails.	Gives a mostly sequenced explanation and a simple prediction.	Gives a list of facts without sequence or without a prediction.
Use of evidence and Kenyan context	Consistently uses the ugali, sukuma wiki and nyama choma meal and practical observations to support claims, with correct scientific vocabulary.	Uses the meal and some observations, with mostly correct vocabulary.	Makes general claims with little reference to the meal or evidence.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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